

Research on Joint Entity-Relation Extraction for the Tibetan "Epic of King Gesar" Based on Multi-Semantic Information Fusion and Global Pointer Network

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Abstract—The "Epic of King Gesar" is a poetic and historical Tibetan literary work and a treasure of Tibetan culture. Guided by the cultural preservation and inheritance of the Tibetan people in China, the paper deeply studies the joint entity-relation extraction method for the "Epic of King Gesar." Addressing issues such as redundant entities, error accumulation from traditional pipeline methods, and insufficient semantic feature capture in Tibetan, the paper proposes an entity-relation extraction method based on multi-semantic information fusion and a global pointer network. (1) CINO was used to obtain dynamic word vectors for the entire text; (2) Global and local features of the Tibetan text were extracted using BiLSTM and CNN; (3) A multi-head attention mechanism was employed to fuse the text's multiple features; (4) The OneRel decoder was used to extract entity relations. Experiments show that this method improves the F1 score by 6.80% compared to the baseline OneRel model, providing a reference for research in specific domains of minority languages in low-resource settings.

Keywords—epic of king gesar, joint entity-relation extraction, multi-semantic information fusion, minority languages, knowledge discovery

I. INTRODUCTION

The Epic of King Gesar is a living epic with numerous versions, widespread influence, a complete structure, and rich poetic expression. It has been passed down orally across the Tibetan Plateau and is hailed as the "Iliad of the East," the "Encyclopedia of Ancient Tibetan Society," and the "Exhibition of National Spiritual Specimens." On September 30, 2009,

UNESCO inscribed it on the Representative List of the Intangible Cultural Heritage of Humanity. As the "culture of their own" for various ethnic groups and regions, intangible cultural heritage plays a vital role in enhancing cultural confidence, promoting national unity, and strengthening the cultural soft power of the nation.

Modern technological means, such as the construction and application of knowledge graphs, offer new possibilities for the protection of intangible cultural heritage. Knowledge graphs can efficiently organize and manage vast amounts of information by representing entities and their relationships, making complex knowledge systems structured, queryable, and analyzable. For epic cultural heritage like *The Epic of King Gesar*, constructing a knowledge graph not only enables a detailed description and analysis of its content, characters, locations, events, and their interrelationships but also promotes the dissemination and sharing of cultural information, thereby enhancing in-depth understanding and research in this domain. Entity relationship extraction, as a key step in constructing knowledge graphs, holds significant research value.

The Epic of King Gesar is the world's longest epic, composed in a combination of prose and poetry, with distinctive ethnic style and dialectal features, reflecting strong local color and oral tradition. In addition, the text has a loose structure and is lengthy, with the head and tail entities of the same triple often scattered across multiple sentences. This results in a higher frequency of nested entities (as shown in Figure 1) and overlapping entity triples. These factors collectively increase the

To effectively address this issue, we first employ a Bi-directional Long Short-Term Memory (BiLSTM) network to capture the contextual dependencies within the text, thereby enhancing the understanding of the interrelationships between entities in a sentence. Next, considering the challenges posed by the lack of clear word boundaries in Tibetan text and the presence of clitic words adhering to entity boundaries, we utilize Convolutional Neural Networks (CNN) to model fine-grained character-level information in the text, effectively addressing the issue of clitic words adhering to entity boundaries. Then, we introduce a Multi-Head Self-Attention (MSA) mechanism to enhance the model's ability to identify critical features, ensuring that important information is not overlooked during the decoding process. Finally, these processed features are fed into the OneRel decoder for final decoding, effectively combining the contextual and relational information of entities to generate accurate extraction results.

II. RELATED WORK

A. Named Entity Recognition (NER)

the National Science and Technology Major Project(No:2022ZD0116100), Tibetan Cultural Inheritance and Protection - Cultural Heritage Protection and Restoration (No:XZ202201ZD0002G), and Tibet University Graduate High-Level Talent Training Program (No:2022-GSP-S108)Identify applicable funding agency here.

B. Relation Extraction

In Chinese and English languages, Wei Z et al. [12] proposed a model called CasRel based on a cascade binary relation extraction framework. It identifies the primary entity in a relation first, and then predicts the corresponding object entity and relation type based on the recognized primary entity. This method improves the accuracy of relation extraction and better handles multiple relations and overlapping entities. Su J et al. [13] employed a global pointer network, GPLinker, which uses a bidirectional global pointer structure and segment representation method to simultaneously annotate entity boundaries and types, as well as identify relationships between entities. This structure optimizes information flow and interdependence between tasks, significantly improving overall extraction task performance. Zheng H et al. [14] introduced a predicate relation-guided PRGC joint extraction mechanism, where the core idea is to make the relationship the central driving force of the extraction process. The model guides entity extraction by identifying relationship trigger words and then recognizes relationships between entities. Although this relation-centered method simplifies the extraction process, it involves multiple sub-modules or steps, leading to exposure bias and limited batch processing capabilities. Shang Y M et al. [15] proposed a single-model framework, OneRel, designed with a specialized scoring mechanism that outputs entities and their relationships through a single forward propagation. For low-resource domains, Li Bin et al. [16] proposed a joint extraction model for ancient Chinese books by injecting domain knowledge into a pre-trained model using prompt learning and combining it with OneRel for joint decoding of entity relations.

III. RESEARCH METHODOLOGY

The proposed model architecture is illustrated in Figure 2. The model is primarily divided into four components: the CINO text vector representation layer, the multi-semantic information extraction layer, the multi-semantic information fusion layer, and the entity-relationship extraction layer. These components collectively enhance the model's extraction performance from multiple perspectives.

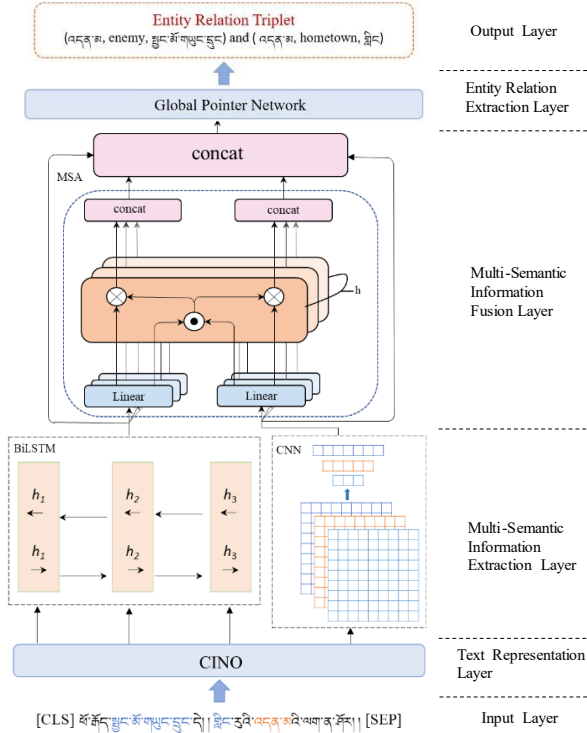


Fig. 2. Model Structure Diagram

A. CINO Multilingual Pre-trained Model

Given the scarcity of language resources for ethnic minorities in China and the limited attention they have received in international research, existing multilingual models exhibit shortcomings when processing texts in these languages. To advance the research and development of information processing for Chinese minority languages, the Harbin Institute of Technology and iFLYTEK Joint Laboratory (HFL) developed the CINO pre-trained model, which has demonstrated outstanding performance on the Wiki-Chinese-Minority (WCM) and other minority language datasets [17]. Therefore, this study employs the CINO-V2-base model to obtain the initial vector representation of the text, thereby improving the understanding and analysis of Tibetan language.

B. BiLSTM Contextual Feature Extraction

Tibetan text is typically complex in structure, with lengthy sentences where relationships between entities often span multiple words, sometimes involving entire sentences. The determination of these relationships depends not only on the semantic content of the entity words themselves but also on the surrounding contextual information. In such cases, recognizing relationships between entities requires consideration of both the

semantics of the entity words and their contextual information. BiLSTM, with its bidirectional structure, can capture these global dependencies through long-term memory, thereby enhancing the model's ability to recognize entity boundaries and types.

BiLSTM (Bidirectional Long Short-Term Memory) is a specialized type of Recurrent Neural Network (RNN) designed to learn long-range dependencies in sequential data analysis. BiLSTM combines two LSTM networks that process the input information along the forward and backward directions of the time series, respectively, and merge the information at each time point to capture both past and future contextual information.

In a BiLSTM network, there are two LSTM chains: one operates in the forward direction $\vec{ht}$, and the other in the backward direction $\overleftarrow{ht}$. The final hidden state is obtained by concatenating the features from both directions, as shown in Equation (1).

$$ht = [\vec{ht}; \overleftarrow{ht}] \quad (1)$$

By using the CINO-encoded vectors as input xt , BiLSTM effectively encodes the contextual information, where each xt represents the output vector corresponding to the respective element in the sequence from CINO.

C. CNN Local Feature Capture Layer

Tibetan differs significantly from languages like Chinese and English, particularly in its agglutinative word characteristics. In Tibetan, many entity names often appear in the form of contracted words, posing challenges in identifying entity boundaries. For instance, in the sentence "ལྷ་མོ་ལྷ་མོ་ལྷ་མོ་ལྷ་མོ་ ལྷ་མོ་ལྷ་མོ་ལྷ་མོ་ལྷ་མོ་" (translation: "On the way back to his mother Gosa's place, Prince Joru wore an antelope skin hat"), the names "ལྷ་མོ་" and "ལྷ་མོ་ལྷ་མོ་" both end with contracted words. Additionally, "ལྷ་མོ་" is an abbreviation composed of "ལྷ་" and the contracted word "མོ་" while "ལྷ་མོ་ལྷ་མོ་" is formed by combining "ལྷ་མོ་" and the contracted word "ལྷ་". Traditional entity recognition methods often struggle to effectively handle entity boundary issues arising from contracted words. Therefore, this paper introduces a Convolutional Neural Network (CNN) to extract character-level local features. CNNs can effectively capture local dependencies and features in text through their convolutional layers, thereby achieving higher accuracy in identifying irregular entity boundaries. When processing Tibetan prose and epic texts, such as *The Epic of King Gesar*, which involve special entities like weapons and mounts, the application of CNNs further addresses the issues of word irregularities caused by text complexity and linguistic specificity [18].

For the sequence vectors output by CINO, this study employs a 3-layer one-dimensional convolution to process them, enabling the model to extract and refine text features at different levels. This provides richer and more discriminative information for subsequent feature fusion and entity-relationship extraction. The specific computation process is as follows:

The first convolution operation (with a kernel size of 3) is described by Equation (2).

$$H_{CNN}^1 = RELU(BN(W_{conv1} * H_{CINO} + b_{conv1})) \quad (2)$$

The second convolution operation (with a kernel size of 5) is described by Equation (3).

$$H_{CNN}^2 = RELU(BN(W_{conv2} * H_{CNN}^1 + b_{conv2})) \quad (3)$$

The third convolution operation (with a kernel size of 7) is described by Equation (4).

$$H_{CNN}^3 = RELU(BN(W_{conv3} * H_{CNN}^2 + b_{conv3})) \quad (4)$$

The final output of the CNN local feature capture layer is H_{CNN}^3 , where H_{CINO} represents the output of CINO, W_{conv1} , W_{conv2} , W_{conv3} are the convolution kernels, $*$ represents the convolution operation, b_{conv1} , b_{conv2} , b_{conv3} are the bias terms, BN denotes batch normalization, and ReLU is used to introduce nonlinearity.

D. Multi-Head Attention Mechanism for Multimodal Information Fusion Layer

The BiLSTM and CNN have already extracted diverse semantic information from the text. Different semantic features carry varying levels of significance for the extraction of entities and relationships in the Gesar domain. To emphasize the more critical features, this paper introduces a multi-head self-attention mechanism, enabling the model to generate different subspaces and reduce the impact of noisy data. The essence of the multi-head self-attention mechanism lies in the multiple applications of the self-attention mechanism, allowing the model to learn relatively important features from different subspaces. Given each input as Query Q, Key K, and Value V, the calculation process for each head in the multi-head attention mechanism is illustrated in Equations (5) ~ (7):

1. Linear Transformation:

$$Q_i = H_{BiLSTM} W_i^Q \quad (5)$$

$$K_i = H_{CNN} W_i^K \quad (6)$$

$$V_i = H_{CNN} W_i^V \quad (7)$$

2. Scaled Dot-Product Attention, as shown in Equation (8):

$$Attention = (Q_i, K_i, V_i) = SoftMax(\frac{Q_i K_i^T}{\sqrt{d_k}}) V_i \quad (8)$$

3. Concatenation of All Heads' Outputs, as shown in Equation (9):

$$MultiHead(Q_i, K_i, V_i) = Concat(head_1, \dots, head_h) W^o \quad (9)$$

where W_i^Q , W_i^K , and W_i^V are the weight matrices for the i -th head, d_k is the dimension of each head, and W^o is the linear transformation matrix. Thus, the model can learn different types of information through the multi-head self-attention mechanism.

E. OneRel Global Pointer Network Decoding Layer

The OneRel decoder constructs a table for each relation, considering the start and end positions of entities from a global perspective, treating the entity head and tail as a whole. This enables the model to seamlessly recognize overlapping relation triplets, as well as nested and non-nested entities. Specifically, the model predicts a set of HB-TB, HB-TE, and HE-TE to capture the complex relational structure between entities and perform entity head-to-tail extraction. (Here, HB-TB represents the start of the head entity to the start of the tail entity, HB-TE represents the start of the head entity to the end of the tail entity, and HE-TE represents the end of the head entity to the end of the tail entity.) This unified extraction of entities and relationships in a single step significantly simplifies the processing flow and reduces the possibility of error propagation. It not only improves the precision of entity and relation extraction but also enhances the model's ability to handle long texts and multi-relation contexts [17].

As shown in Fig. 3, the decoder first segments the text according to Tibetan characters, constructing a matrix with both width and height equal to the input text length, where each Tibetan character corresponds to the respective row and column. For example, consider the sentence "པོ་རྒྱལ་ལྷ་མོ་གཞུང་དུ་དེ། རྒྱལ་པོ་འདུན་མའི་ལག་ན་ཤོར།" (translated as: "The brave warrior Jangmu Yongzhen died at the hands of Ling people's Danma"), which contains overlapping relation triplets (འདུན་མ, enemy, རྒྱལ་པོ་གཞུང་དུ་དེ) and (འདུན་མ, hometown, རྒྱལ་པོ). In the "enemy" relation matrix, "འདུན" and "རྒྱལ" are marked as "HB-TB," "འདུན" and "དེ" are marked as "HB-TE," and "མ" and "དེ" are marked as "HE-TE." This uniquely determines the entity boundaries under the current relation, allowing for accurate extraction of the triplet (འདུན་མ, enemy, རྒྱལ་པོ་གཞུང་དུ་དེ). Similarly, other entity relations are extracted in the same way.

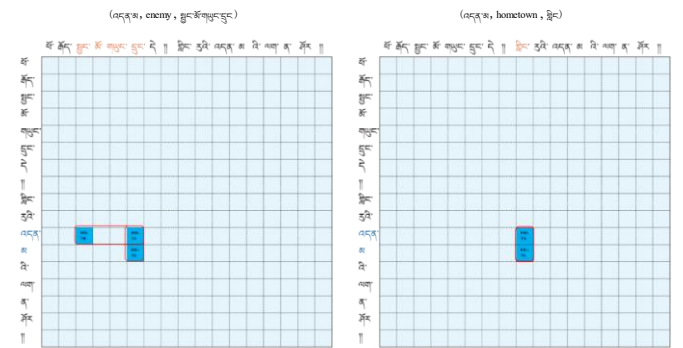


Fig. 3. OneRel Global Pointer Network Decoder

The specific calculation process is as follows: After obtaining the comprehensive features of each entity pair through the multi-headed attention mechanism, for all possible word pairs (e_i, e_j) in the sentence and the predefined relation r_k , a classifier is used to assign high-confidence labels. The classifier's input is the combination of entity pairs and relations (e_i, r_k, e_j) , and it uses a nonlinear mapping function to compute, as shown in Equation (10):

$$h \otimes t = \phi(W[h; t]^T + b) \quad (10)$$

where W and b are trainable weights and biases, $[\cdot]$ is the concatenation operation, and $RELU$ is the nonlinear activation function used in this paper.

Next, the relation embedding R is used to score each label category, as shown in Equation (11):

$$v(w_i, w_k, w_j)_{k=1}^K = R^T \phi(drop(W[e_i; e_j]^T + b)) \quad (11)$$

where v is the score vector, and $drop(\bullet)$ denotes the application of the dropout strategy to prevent overfitting.

Finally, the score vector is converted into probabilities through the *softmax* function, and the model predicts the label of each triplet based on these probabilities, determining whether the triplet is valid, as shown in Equation (12):

$$P(y(w_i, r_k, w_j)|S) = Soft\ max(v(w_i, r_k, w_j)) \quad (12)$$

IV. RESULTS AND ANALYSIS

A. Dataset Construction

The corpus used in this study was provided by the Gesar Research Office of the China National Unity Research Institute

at Tibet University. It consists of the Tibetan version of *The Epic of King Gesar* as narrated by the renowned Tibetan storyteller, Zhaba, which is the longest and most content-rich oral epic in human literature to date. *The Epic of King Gesar* holds immense value for Tibetan studies. Upon careful recommendation by researchers at the Gesar Research Office, this study selected four representative works: *The Magical King of Bagar*, *The Battle of Menling*, *The Conquest of the Northern Demon King*, and *The Jewel Dharma of Zhang Zhung*. The original corpus was not available in electronic form, so the paper texts were first converted to digital documents using scanning and recognition technology. The electronic documents were then carefully proofread against the original paper texts to ensure accuracy. After these steps, a complete electronic document was formed, which was used as the source data to define the annotation standards for entities and relationships. Professional annotators were then organized to manually annotate the data. The final dataset consists of 2,720 annotated sentences, totaling 5,118 triples, encompassing 7 types of entities and 18 types of entity relationships. The dataset was split into training, validation, and test sets in an 8:1:1 ratio. Among them, some types of entity relationships in the triplets are shown in Table 1.

TABLE I. PARTIAL TRIPLET ENTITY-RELATION EXAMPLES

Subject Type	Relation Type	Object Type	Example Triplet	Meaning of Triplet
Person Name	Enemy	Person Name	(གེ་སར་རྒྱལ་པོ་, Enemy, ཀ་ལ་འབྲུག་གི་རྒྱལ་པོ་) (King Gesar, Enemy, Kalar's Trickster King)	Indicates a conflict between the subject and object, such as a fight between two individuals.
Person Name	General	Person Name	(གེ་སར་རྒྱལ་པོ་, General, ཚ་ཁང་འདད་ན་མ་) (King Gesar, General, Chashang Danma)	The subject is a king, and the object refers to the generals and warriors under the king's command.
Person Name	Weapon	Weapon	(རྩ་མཁའ་དབྱུ་མ་, Weapon, རལ་གྱི་ཡར་ཤི་རྩ་མ་མཁའ་) (Prince Zara, Weapon, Yas Meteor Sword)	The subject possesses a weapon, such as armor, helmets, arrows, swords, etc.
Person Name	Mount	Mount	(རྩ་མ་ཁའ་དགའ་, Mount, རྩ་མ་ཁའ་དགའ་) (Jiacha Xiega, Mount, Jiaxiasuga)	The subject's exclusive mount, such as a horse or lion.
Person Name	Song	Song	(འབྲུག་མོ་, Song, དལ་མེད་བྱུག་འབྲུར་) (Drukmo, Song, Nine-Man Six-Transformation Song)	The object is the tune or song title used by the subject during performance.

B. Evaluation Metrics

In this experiment, precision (P), recall (R), and F1 score (F1) were used as evaluation metrics for the joint extraction of entities and relationships. Specifically, if both the entity types and relationship types in a predicted triple are correct, the triple is considered correctly predicted; otherwise, if the entity type of a triple is predicted incorrectly, the entire triple is considered incorrect. Here, TP denotes the number of correctly predicted entity relationships, FP denotes the number of incorrectly identified entity relationships, and FN denotes the number of entity relationships that were not identified. The specific calculations are shown in Equations (13) to (15).

$$P = \frac{TP}{TP+FP} \quad (13)$$

$$R = \frac{TP}{TP+FN} \quad (14)$$

$$F1 = \frac{2 \cdot P \cdot R}{P+R} \quad (15)$$

C. Experimental Setup

The experimental setup for this study is configured as follows: The operating system is Windows 10 Workstation Edition, the CPU is an Intel(R) Xeon(R) Gold 6138 CPU @ 2.00 GHz with dual processors, and the graphics card is a GTX 2080 TI with 22 GB of video memory. The deep learning framework employed is PyTorch 2.2.2 with CUDA 11.8 support.

For model training, the hyperparameters were set as follows: The size of the LSTM hidden layer was set to 128, and the

number of LSTM layers was 1. In the Convolutional Neural Network (CNN) section, we utilized convolutional kernels of sizes 3, 5, and 7, with a total of 3 layers in the CNN. In the Multi-Head Self-Attention (MSA) mechanism, we configured 8 attention heads. To mitigate the risk of model overfitting, a dropout rate of 0.1 was introduced into the network. The learning rate for model training was set to $2e-5$, and the maximum length of the input sequence was restricted to 300 units.

D. Result Analysis

To verify the effectiveness of the proposed method, it was compared with four advanced baseline models for joint extraction performance. The experimental results demonstrate that the proposed model significantly outperforms the baseline models. The experimental results are shown in Table 2.

TABLE II. COMPARISON OF EXPERIMENTAL RESULTS FOR THE JOINT EXTRACTION MODEL (%)

Models	P	R	F1
CasRel	67.77	41.92	49.91
GPLinker	77.09	53.08	62.87
PRGC	65.35	59.36	62.21
OneRel	63.02	63.15	63.08
Ours	73.83	66.33	69.88

Based on the bar chart in Figure 4, it is evident that the proposed model outperforms others in terms of recall and F1 score, showing improvements of 3.18% and 6.80% respectively over the baseline model OneRel. This indicates that the multi-semantic feature fusion strategy effectively enhances the accuracy and comprehensiveness of entity and relationship extraction. The combination of BiLSTM and CNN optimizes the deep feature extraction of text, while the multi-head attention mechanism effectively integrates these features, improving the model's ability to handle the mixed-text structure of epic prose.

Although the CasRel model demonstrates excellent precision, its recall and F1 scores are relatively low, suggesting that while CasRel is capable of correctly identifying some entity relationships, it has a significant issue with missed detections. This may be due to its limitations in handling complex or nested relationships, leading to incomplete extraction of relevant entities and relationships, resulting in suboptimal overall performance. In contrast, the GPLinker model outperforms CasRel across all metrics, with notable improvements in precision and F1 score, reaching 77.09% and 62.87%, respectively. Despite its recall rate being 53.08%, it still exhibits a relatively balanced performance. This suggests that GPLinker provides a more comprehensive identification of entities and relationships, likely due to its design that combines global and local pointers, improving the extraction of inter-entity relationships. The PRGC model achieves a higher recall rate, nearly 60%, indicating its outstanding performance in detecting a broader range of entity relationships. This suggests that its iterative strategy of prediction, correction, and confirmation enhances the precision and comprehensiveness of relationship extraction. However, its precision is slightly lower than that of GPLinker, decreasing by 11.74%, and its F1 score drops by 0.66%, highlighting an issue with a higher false positive rate. The OneRel model performs relatively balanced across all

metrics, which may be attributed to its simplified single relationship label strategy, which streamlines the model structure, maintaining a reasonable level of precision while also achieving a relatively high recall rate. However, this simplification might limit the model's ability to handle more complex relationships.

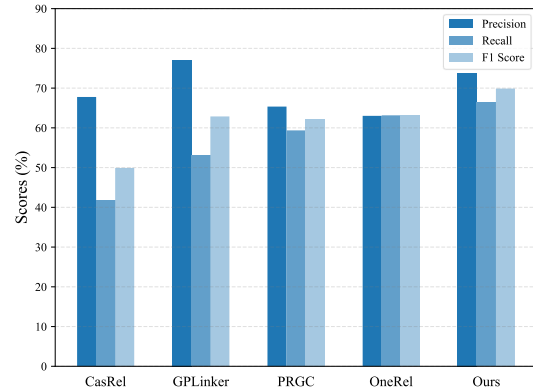


Fig. 4. Comparative Experiments

E. Ablation Study

To verify the contribution of each module proposed in this study for joint entity-relation extraction in Tibetan texts, ablation experiments were conducted by systematically removing each module. The results are presented in Table 3.

TABLE III. ABLATION EXPERIMENT RESULTS FOR THE JOINT EXTRACTION OF TIBETAN GESAR (%)

Models	P	R	F1
Ours	73.83	66.33	69.88
-BiLSTM	69.53	64.54	66.94
-CNN	69.71	62.35	65.83
-BiLSTM,MSA	67.38	62.55	64.88
-CNN,MSA	68.78	62.75	65.62

As shown in the bar chart in Figure 5, it is evident that the combination of BiLSTM, CNN, and Multi-Head Self-Attention (MSA) provides the best performance. This indicates that the integration of these three techniques optimizes the model's overall capability in handling entity-relation extraction tasks. BiLSTM captures contextual dependencies, CNN enhances local feature recognition, and MSA effectively integrates these features for precise relation extraction.

When BiLSTM was removed, the model's performance declined across all metrics. BiLSTM is crucial for capturing long-distance dependencies, especially when complex semantic relationships implicit in the text need to be understood within the entire sentence context. Its absence led to a loss in the model's ability to comprehend the text, particularly when dealing with prose poetry that has intricate structures and semantics.

Upon removing the CNN, the model's precision decreased by 4.12%, recall by 3.98%, and F1 score by 4.05%, indicating a weakened ability to capture all relevant entities and relations. CNN is vital for identifying key local features such as entity

boundaries and local semantic patterns; its absence reduced the model's sensitivity to localized regions of the text.

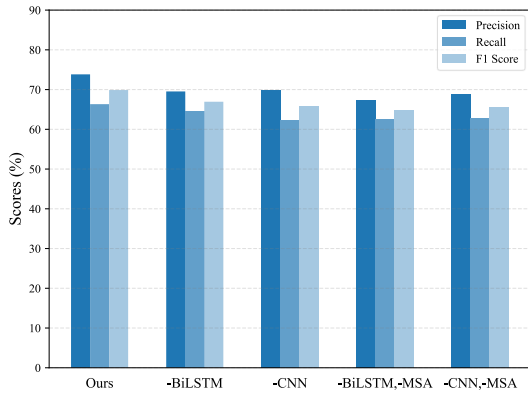


Fig. 5. Ablation Experiments

The model's performance significantly deteriorated when both BiLSTM and MSA were removed. The removal of BiLSTM reduced the model's deep understanding of the text's context, while the absence of MSA diminished the model's ability to effectively integrate diverse features. The combined lack of these components further amplified the decline in performance.

Similarly, the performance drop was also notable when CNN and MSA were removed, underscoring the positive roles that CNN and MSA play in feature capture and fusion. Removing CNN reduced the ability to capture local details, while the absence of MSA weakened the model's capability to extract the most useful information from these local features, leading to a loss in overall precision for complex relation extraction.

V. CONCLUSION

In this paper, we proposed a method for joint entity and relation extraction in the Tibetan *Epic of King Gesar* by leveraging a pre-trained model specifically designed for minority languages to obtain word embeddings, thereby capturing richer semantic features. BiLSTM was employed to capture contextual information and global dependencies, followed by CNN to capture local features of the text, addressing challenges related to entity boundaries and agglutinative words in Tibetan, thus extracting more fine-grained semantic features. Finally, a multi-head self-attention mechanism was used to capture crucial textual features, enhancing the model's information extraction capabilities, with triples being extracted via OneRel decoding. Comparative experiments and ablation studies on the Tibetan dataset demonstrate the effectiveness of the proposed method.

Future work will focus on expanding the Tibetan *Epic of King Gesar* dataset to develop a comprehensive Gesar database and integrating graph database technology to construct a Gesar knowledge graph. We will also explore the combination of large models and prompt engineering techniques to assist in information extraction, aiming to further improve the efficiency and accuracy of the extraction process.

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